
DETERMINANT

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1 Multiplicative Property

In this section we assume the usual definition of a determinant, from which one can derive the following properties that we will need:

$$\det \text{diag}(d_1, d_2, \dots, d_n) = d_1 d_2 \dots d_n, \quad (1)$$

$$\det AB = \det A \det B, \quad (2)$$

$$\det \lambda A = \lambda^n \det A. \quad (3)$$

Now our task is to find all functions $f(A)$ from matrices to real numbers that satisfy the multiplicative property:

$$f(AB) = f(A)f(B). \quad (4)$$

We will use (4) repeatedly in the following calculation. First we decompose the matrix $A = HU$ using a polar decomposition, where H is Hermitian and U is unitary. Both H and U are individually diagonalizable (but not mutually diagonalizable, since A is not in general diagonalizable). For any matrix C that is diagonalizable, we can write $C = PDP^{-1}$ where D is diagonal and P is invertible. Then:

$$f(C) = f(PDP^{-1}) = f(P)f(D)f(P^{-1}) = f(P)f(P^{-1})f(D) = \quad (5)$$

$$= f(PP^{-1}D) = f(D). \quad (6)$$

That means that if a matrix is diagonalizable, then f applied to that matrix is equal to f applied to the diagonal matrix.

Below we will use matrices P_i that are unit matrices with the 1st and i -th rows permuted. Now we can compute, using the fact that H and U can be decomposed into diagonal matrices D_H and D_U , and their product is also a diagonal matrix $D = D_H D_U$:

$$f(A) = f(HU) = f(H)f(U) = f(D_H)f(D_U) = f(D_H D_U) = f(D) = \quad (7)$$

$$= f(\text{diag}(d_1, 1, 1, \dots))f(\text{diag}(1, d_2, 1, \dots)) \dots f(\text{diag}(1, \dots, 1, d_n)) = \quad (8)$$

$$= f(P_1 \text{diag}(d_1, 1, 1, \dots) P_1^{-1}) f(P_2 \text{diag}(d_2, 1, 1, \dots) P_2^{-1}) \dots \quad (9)$$

$$\dots f(P_n \text{diag}(d_n, 1, 1, \dots) P_n^{-1}) = \quad (10)$$

$$= f(\text{diag}(d_1, 1, 1, \dots) \text{diag}(d_2, 1, 1, \dots) \dots \text{diag}(d_n, 1, 1, \dots)) = \quad (11)$$

$$= f(\text{diag}(d_1 d_2 \dots d_n, 1, 1, \dots)) = \quad (12)$$

$$= f(\text{diag}(\det D, 1, 1, \dots)) = \quad (13)$$

$$= g(\det D) = g(\det D_H D_U) = g(\det D_H \det D_U) = g(\det H \det U) = \quad (14)$$

$$= g(\det HU) = g(\det A), \quad (15)$$

where we defined a function $g(x)$ using

$$g(x) = f(\text{diag}(x, 1, 1, \dots)) . \quad (16)$$

Let's compute the functional equation for $g(x)$:

$$g(xy) = f(\text{diag}(xy, 1, 1, \dots)) = \quad (17)$$

$$= f(\text{diag}(x, 1, 1, \dots) \text{diag}(y, 1, 1, \dots)) = \quad (18)$$

$$= f(\text{diag}(x, 1, 1, \dots)) f(\text{diag}(y, 1, 1, \dots)) = \quad (19)$$

$$= g(x)g(y) . \quad (20)$$

In the above computations we have only used (4) and no other assumptions.

Let's summarize our results. From (4) it follows that:

$$f(A) = g(\det A) , \quad (21)$$

$$g(xy) = g(x)g(y) , \quad (22)$$

for numbers x, y .

The equation (22) can be solved in several different ways.

Approach 1: Assuming that $f(A)$ (and thus $g(x)$) is measurable, we show below that the only solutions are $g(x) = 0$, $g(x) = |\text{sign}(x)|$, $g(x) = |x|^s$ and $g(x) = \text{sign}(x)|x|^s$. Consequently the only nonzero solution to (4) is a power of a determinant (optionally multiplied with the signum function). If $f(A)$ is not measurable then we also get highly pathological solutions (see the section below). The above is thus the most general solution.

Approach 2: Instead of assuming measurability, we can instead assume homogeneity:

$$f(\lambda A) = \lambda^n f(A) , \quad (23)$$

where n is the (integer) dimension of the matrix A . Using (21) and (22) we get:

$$f(\lambda A) = g(\det \lambda A) = g(\lambda^n \det A) = g(\lambda^n)g(\det A) = g(\lambda^n)f(A) . \quad (24)$$

Comparing to (23) we get:

$$g(\lambda^n) = \lambda^n . \quad (25)$$

Since λ can be any real (or complex) number, we get:

$$g(x) = x \quad (26)$$

and from (21) we get:

$$f(A) = \det A . \quad (27)$$

Summary:

- Assuming equations (4) and (23) we obtained (27) as the only nonzero solution. That means that we can define a determinant using (4) and (23) and no other assumptions.
- Alternatively, if we only assume (4) and that $f(A)$ is measurable, then the only nonzero solution is a power of a determinant (optionally multiplied with the signum function).

TODO

The transition from λ^n to x in (25) might require better treatment, there might be some hidden assumptions about continuity on $g(x)$. Maybe there is a way to derive $g(x) = x$ more directly. We can maybe use $g(\lambda^n) = g(\lambda)^n = \lambda^n$. For $n = 2$, we get: $g(\lambda)^2 = \lambda^2$, which also has a solution $g(x) = |x|$. But if we allow λ to be complex, then likely the only solution is $g(x) = x$ without any additional assumptions.

2 Cauchy's Additive Functional Equation

The Cauchy's functional equation ([Wikipedia \(2025\)](#)) usually means the additive equation of this form:

$$f(x + y) = f(x) + f(y). \quad (28)$$

There are other related equations, such as the Cauchy's multiplicative functional equation in the next section.

For integer n we get:

$$f(nx) = f(\underbrace{x + x + \cdots + x}_n) = \underbrace{f(x) + f(x) + \cdots + f(x)}_n = nf(x). \quad (29)$$

By substituting $y = nx$ we get $f(y) = nf(y/n)$, from which $f(y/n) = (1/n)f(y)$. Using these two relations for multiplication and division by an integer we obtain for any rational number $q = \frac{p}{q}$, where p, q are integers:

$$f(qx) = f\left(\frac{p}{q}x\right) = pf\left(\frac{x}{q}\right) = \frac{p}{q}f(x) = qf(x). \quad (30)$$

We also have to prove it for $p \leq 0$ which we can use $f(0) = f(0) + f(0)$ which implies $f(0) = 0$ and using $0 = f(0) = f(x - x) = f(x) + f(-x)$ from which follows $f(-x) = -f(x)$. Setting $x = 1$ we get for all rational q :

$$f(q) = qf(1) = cq. \quad (31)$$

where $c = f(1)$ is any real constant. If we assume that $f(x)$ is continuous, then this implies that $f(x) = cx$ for all real x , since the rational numbers are dense in real numbers. It turns out that one can relax the continuity requirement to only require that $f(x)$ is measurable. Since $f(x) = cx$ is linear, we get:

Theorem

Additive, measurable functions are linear.

A linear function $f(x) = ax + b$ can be substituted into (28) and one obtains that $b = 0$. So this theorem is a simple way to remember the solutions to the Cauchy's additive functional equation.

Proof

To prove this theorem, we introduce $g(x) = f(x) - f(1)x$ and prove that $g(x) = 0$, from which it follows that $f(x) = f(1)x = cx$. We have:

$$g(x + y) = f(x + y) - f(1)(x + y) = f(x) - f(1)x + f(y) - f(1)y = g(x) + g(y), \quad (32)$$

so $g(x)$ is also additive. For rational q we have:

$$g(q) = f(q) - f(1)q = cq - f(1)q = 0. \quad (33)$$

Also we get:

$$g(x + q) = g(x) + g(q) = g(x), \quad (34)$$

so $g(x)$ is periodic with a period q . It can be shown from this periodicity and function $g(x)$ being measurable that $g(x)$ is essentially bounded. Now we can compute the following integral which exists and is finite:

$$\int_0^1 g(x)dx = \int_0^1 g(x + y)dx = \int_0^1 g(x)dx + \int_0^1 g(y)dx = \int_0^1 g(x)dx + g(y), \quad (35)$$

so $g(y) = 0$ for all real y . Here we have used the dominated convergence theorem for $g(x) = g(x + q_n)$ where a sequence of rational q_n is converging towards the real y .

Note

If we do not require a measurable function, then there are infinitely many solutions to (28) that are nonlinear and their graphs are dense. They are highly pathological functions and these solutions are all excluded by requiring that $f(x)$ is measurable.

3 Cauchy's Multiplicative Functional Equation

The Cauchy's multiplicative functional equation is:

$$f(xy) = f(x)f(y). \quad (36)$$

There are only 3 cases that can happen:

- $f(x) = 0$.
- $f(x) = 1$.
- $f(x) = \begin{cases} h(x) & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ \pm h(|x|) & \text{if } x < 0. \end{cases}$

In the last case the function $h(x)$ is positive $h(x) > 0$ and defined only for $x > 0$. The $x < 0$ case has two options for even and odd $f(x)$. We can equivalently write the two cases explicitly as follows:

- Even: $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ h(|x|) & \text{if } x \neq 0. \end{cases}$
- Odd: $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ \text{sign}(x)h(|x|) & \text{if } x \neq 0. \end{cases}$

As a special case for $h(x) = 1$ we get:

- Even: $f(x) = |\text{sign}(x)| = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \neq 0. \end{cases}$
- Odd: $f(x) = \text{sign}(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x > 0, \\ -1 & \text{if } x < 0. \end{cases}$

We can then use the substitution $g(x) = \log f(e^x)$ to convert the multiplicative equation into an additive equation from the previous section:

$$g(x+y) = \log f(e^{x+y}) = \log f(e^x e^y) = \log(f(e^x)f(e^y)) = \quad (37)$$

$$= \log f(e^x) + \log f(e^y) = g(x) + g(y). \quad (38)$$

We find the solution using the previous section, and then we compute $h(x)$ using $h(x) = \exp(g(\log x))$, which satisfies:

$$h(xy) = \exp(g(\log(xy))) = \exp(g(\log x + \log y)) = \exp(g(\log x) + g(\log y)) = \quad (39)$$

$$= \exp(g(\log x)) \exp(g(\log y)) = h(x)h(y). \quad (40)$$

For measurable $g(x)$ the solution to the additive equation is $g(x) = cx$, so we assume $f(x)$ is measurable (which implies $g(x)$ is measurable) and we get $h(x) = \exp(c \log x) = \exp(\log(x^c)) = x^c$ and the four solutions are

- $f(x) = 0$.
- $f(x) = 1$.
- $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ h(|x|) & \text{if } x \neq 0, \end{cases} = \begin{cases} 0 & \text{if } x = 0, \\ |x|^c & \text{if } x \neq 0. \end{cases}$
- $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ \text{sign}(x)h(|x|) & \text{if } x \neq 0, \end{cases} = \begin{cases} 0 & \text{if } x = 0, \\ \text{sign}(x)|x|^c & \text{if } x \neq 0. \end{cases}$

The solutions can be equivalently written as:

- $f(x) = 0$.
- $f(x) = |\text{sign}(x)| = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \neq 0. \end{cases}$
- $f(x) = |x|^c$, with $f(0) = 0$ (needed for $c < 0$).
- $f(x) = \text{sign}(x)|x|^c$, with $f(0) = 0$ (needed for $c < 0$).

The solution $f(x) = 1$ is already included in $|x|^c$ for $c = 0$. If c is an integer, then the last two solutions can always be written (for both even and odd integers) as $f(x) = x^c$ and $f(x) = \text{sign}(x)x^c$.

Some examples are: $0, 1, \text{sign}(x), |\text{sign}(x)|, x, |x|, x^2, \text{sign}(x)x^2, x^3, |x|^3, \frac{1}{x}, \frac{1}{|x|}, \frac{1}{x^2}, \frac{\text{sign}(x)}{x^2} = \frac{|x|}{x^3}, \frac{1}{x^3}, \sqrt{|x|}, \frac{1}{\sqrt{|x|^3}}$, etc.

Note

The function $h(x)$ is only defined for positive x and that is all we need. But if we are curious how the formula evaluates for $x \leq 0$, we get the following. For $c > 0$ and $x = 0^+$ we get:

$$h(x) = \exp(c \log 0^+) = \exp(-c \infty) = 0. \quad (41)$$

For $x < 0$ we get:

$$h(x) = \exp(c(\log |x| + i \arg x)) = \exp(c(\log(-x) + i\pi)) = \quad (42)$$

$$= \exp(\log((-x)^c)) \exp(ic\pi) = (-x)^c \exp(ic\pi). \quad (43)$$

For an even integer c we have $\exp(ic\pi) = 1$ and $h(x) = (-x)^c = x^c$, for an odd integer c we have $\exp(ic\pi) = -1$ and $h(x) = -(-x)^c = x^c$ and for non-integer c we get some complex number. For example, for $c = 1$ (which is an odd integer) we get:

$$h(x) = (-x)(-1) = x. \quad (44)$$

Due to the discontinuity at $x = 0$, we can stitch solutions there, and extend $h(x)$ as even or odd functions.

References

Wikipedia. Cauchy's functional equation. <https://en.wikipedia.org/wiki/Cauchy2025>. URL https://en.wikipedia.org/wiki/Cauchy%27s_functional_equation. [Online; accessed 2025-01-17].